

Design of Shaft

Shaft is a common and important machine element. It is a rotating member, in general, has a circular cross-section and is used to transmit power. The shaft may be hollow or solid. The shaft is supported on bearings and it rotates a set of gears or pulleys for the purpose of power transmission. The shaft is generally acted upon by bending moment, torsion and axial force. Design of shaft primarily involves in determining stresses at critical point in the shaft that is arising due to aforementioned loading. Other two similar forms of a shaft are axle and spindle.

Axle is a non-rotating member used for supporting rotating wheels etc. and do not transmit any torque. Spindle is simply defined as a short shaft. However, design method remains the same for axle and spindle as that for a shaft.

Standard sizes of Shafts

Typical sizes of solid shaft that are available in the market are,

Up to 25 mm 0.5 mm increments

25 to 50 mm 1.0 mm increments

50 to 100 mm 2.0 mm increments

100 to 200 mm 5.0 mm increments **or refer Table 3.5(a)/Pg.57**

Material Used for Shafts

The material used for shafts should have the following properties:

1. It should have high strength.
2. It should have good machinability.
3. It should have low notch sensitivity factor.
4. It should have good heat treatment properties.
5. It should have high wear resistant properties.

The material used for ordinary shafts is carbon steel of grades 40 C 8, 45 C 8, 50 C 4 and 50 C 12.

Material for Shafts

The ferrous, non-ferrous materials and nonmetals are used as shaft material depending on the application. Some of the common ferrous materials used for shaft are discussed below.

Hot-rolled plain carbon steel

These materials are least expensive. Since it is hot rolled, scaling is always present on the surface and machining is required to make the surface smooth.

Cold-drawn plain carbon/alloy composition

Since it is cold drawn it has got its inherent characteristics of smooth bright finish. Amount of machining therefore is minimal. Better yield strength is also obtained. This is widely used for general purpose transmission shaft.

Alloy steels

Alloy steel as one can understand is a mixture of various elements with the parent steel to improve certain physical properties. To retain the total advantage of alloying materials one requires heat treatment of the machine components after it has been manufactured. Nickel, chromium and vanadium are some of the common alloying materials. However, alloy steel is expensive.

These materials are used for relatively severe service conditions. When the situation demands great strength then alloy steels are used. They have fewer tendencies to crack, warp or distort in heat treatment. Residual stresses are also less compared to CS (Carbon Steel).

In certain cases the shaft needs to be wear resistant, and then more attention has to be paid to make the surface of the shaft to be wear resistant. *The common types of surface hardening methods are,*

Hardening of surface

Case hardening and carburizing

Cyaniding and nitriding

Design considerations for shaft

For the design of shaft following two methods are adopted,

Design based on Strength

In this method, design is carried out so that stress at any location of the shaft should not exceed the material yield stress. However, no consideration for shaft deflection and shaft twist is included.

Design based on Stiffness

Basic idea of design in such case depends on the allowable deflection and twist of the shaft.

Design based on Strength

The stress at any point on the shaft depends on the nature of load acting on it. The stresses which may be present are as follows.

Basic stress equations:

- (a) Shafts subjected to twisting moment or torque only,
- (b) Shafts subjected to bending moment only,
- (c) Shafts subjected to combined twisting and bending moments, and
- (d) Shafts subjected to axial loads in addition to combined torsional and bending loads.

Maximum Permissible Working Stresses for Transmission Shafts

According to American Society of Mechanical Engineers (ASME) code for the design of transmission shafts, the maximum *permissible working stresses in tension or compression* may be taken as **(Table 3.5(b)/Pg.57**

(a) 110 MPa for shafts without allowance for keyways.

(b) 83MPa for shafts with allowance for keyways.

For shafts purchased under definite physical specifications, the permissible tensile stress may be taken as 60 per cent of the elastic limit in tension, but not more than 36 per cent of the ultimate tensile strength

The maximum permissible shear stress may be taken as

(a) 55 MPa for shafts without allowance for key ways.

(b) 41 MPa for shafts with allowance for keyways.

For shafts purchased under definite physical specifications, the permissible shear stress (τ) may be taken as 30 per cent of the elastic limit in tension but not more than 18 per cent of the ultimate tensile strength.

For combined stress

(a) 55 MPa for shafts without allowance for keyways.

(b) 41MPa for shafts with allowance for keyways.

For shafts purchased under definite physical specifications (without keyways*), the permissible shear stress (τ) may be taken as 30 per cent of the elastic limit in tension but not more than 18 per cent of the ultimate tensile strength.

*** The values are to be reduced by 25 % if keyways are present**

Maximum shear stress theory

Design of the shaft mostly uses maximum shear stress theory. It states that a machine member fails when the maximum shear stress at a point exceeds the maximum allowable shear stress for the shaft material.

Equivalent twisting moment (T_{eq})

The equivalent twisting moment may be defined as that twisting moment, which when acting alone, produces the same shear stress (τ) as the actual twisting moment and bending moment acting together.

Maximum shear stress theory

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_b}{2}\right)^2 + \tau^2}$$

For circular shaft with diameter d and subjected with bending moment (M) the bending stress is

$$\sigma_b = \frac{32M}{\pi d^3}$$

For circular shaft with diameter d and subjected with torsional moment (T) the shear stress is

$$\tau = \frac{16T}{\pi d^3}$$

$$\tau_{\max} = \sqrt{\left(\frac{\sigma_b}{2}\right)^2 + \tau^2} = \sqrt{\left(\frac{32M}{2 \times \pi d^3}\right)^2 + \left(\frac{16T}{\pi d^3}\right)^2} = \frac{16}{\pi d^3} \sqrt{M^2 + T^2} = \frac{16T_{\text{eq}}}{\pi d^3}$$

$$\text{Where } T_{\text{eq}} = \text{Equivalent torsional moment} = \sqrt{M^2 + T^2}$$

The equivalent bending moment (M_{eq}) may be defined as that moment which when acting alone produces the same tensile or compressive stress (σ_b) as the actual bending moment and twisting moment acting together

Maximum normal stress theory

$$\sigma_{\max} = \frac{\sigma_b}{2} + \sqrt{\left(\frac{\sigma_b}{2}\right)^2 + \tau^2} = \frac{32M}{2 \times \pi d^3} + \sqrt{\left(\frac{32M}{2 \times \pi d^3}\right)^2 + \left(\frac{16T}{\pi d^3}\right)^2} = \frac{32}{\pi d^3} \left[\frac{1}{2} (M + \sqrt{M^2 + T^2}) \right]$$
$$\sigma_{\max} = \frac{32M_{\text{eq}}}{\pi d^3}$$

$$\text{Where } M_{\text{eq}} = \text{Equivalent bending moment} = \left[\frac{1}{2} (M + \sqrt{M^2 + T^2}) \right]$$

Shaft Design based on Strength

1. Shaft subjected to Torque (T) only

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{L} \quad \text{Eqn 1.3(b)/Pg.3}$$

a. Hollow shaft

Eqn 3.4(a)/Pg.50

$$\text{Outside diameter} = d_o, \text{ Inside diameter} = d_i, K = \frac{d_i}{d_o}$$

$$d_o = \left[\frac{16T}{\pi\tau_d} \left(\frac{1}{1-K^4} \right) \right]^{\frac{1}{3}}$$

b. Solid shaft, $K = 0$

$$d_o = d = \left[\frac{16T}{\pi\tau_d} \right]^{\frac{1}{3}}$$

2. Shaft subjected to Bending Moment (M) only

$$\frac{M}{I} = \frac{\sigma}{c} = \frac{E}{R} \quad \text{Eqn 1.3(a)/Pg.3}$$

a. Hollow shaft

Eqn 3.4(b)/Pg.50

Outside diameter = d_o , Inside diameter = d_i , $K = \frac{d_i}{d_o}$

$$d_o = \left[\frac{32M}{\pi\sigma_d} \left(\frac{1}{1-K^4} \right) \right]^{\frac{1}{3}}$$

b. Solid shaft, $K = 0$

$$d_o = d = \left[\frac{32M}{\pi\sigma_d} \right]^{\frac{1}{3}}$$

3. Shaft subjected to combined Bending Moment (M) and Torque (T)

a. Maximum normal stress theory

Eqn 3.5(a)/Pg.50

$$d_o = \left[\frac{16}{\pi\sigma_{\max}} \left(M + \sqrt{M^2 + T^2} \right) \times \left(\frac{1}{1-K^4} \right) \right]^{\frac{1}{3}}$$

b. Maximum shear stress theory

Eqn 3.5(b)/Pg.50

$$d_o = \left[\frac{16}{\pi\tau_{\max}} \left(\sqrt{M^2 + T^2} \right) \times \left(\frac{1}{1-K^4} \right) \right]^{\frac{1}{3}}$$

ASME design Code (Shock and Endurance Factors)

Table 3.1/Pg.56

It was assumed that the shaft is subjected to constant torque and bending moment. But in actual

practice, the shafts are subjected to fluctuating torque and bending moments. In order to design such shafts like line shafts and counter shafts, the combined shock and fatigue factors must be taken into account for the computed twisting moment (T) and bending moment (M). These factors are C_m for bending and C_t shaft for torsion.

a. Maximum normal stress theory

Eqn 3.6(a)/Pg.51

$$d_o = \left[\frac{16}{\pi \sigma_{\max}} \left(C_m M + \sqrt{(C_m M)^2 + (C_t T)^2} \right) \times \left(\frac{1}{1 - K^4} \right) \right]^{\frac{1}{3}}$$

b. Maximum shear stress theory

Eqn3.6 (b)/Pg.51

$$d_o = \left[\frac{16}{\pi \tau_{\max}} \left(\sqrt{(C_m M)^2 + (C_t T)^2} \right) \times \left(\frac{1}{1 - K^4} \right) \right]^{\frac{1}{3}}$$

The outside diameter of hollow shaft subjected to an axial load F in addition to the bending and torsional loads.

Eqn3.8 (b)/Pg.51

$$d_o = \left[\frac{16}{\pi \tau_{\max}} \left(\sqrt{\left(C_m M + \frac{\alpha F d_o (1 + K^2)}{8} \right)^2 + (C_t T)^2} \right) \times \left(\frac{1}{1 - K^4} \right) \right]^{\frac{1}{3}}$$

Where α = ratio of maximum intensity of stress resulting from the axial load to the average axial stress from Table 3.2/Pg.56

$$\alpha = \frac{1}{1 - 0.0044 \left(\frac{L}{k} \right)} \quad \text{When } L/k < 115$$

$$\alpha = \frac{\sigma_y}{n \pi^2 E} (L/k)^2 \quad \text{When } L/k > 115$$

Where L = Length between supporting bearings, mm

k = Radius of gyration of the shaft, mm

σ_y = yield stress in compression, N/mm²

E = Modulus of Elasticity, N/mm²

n = Constant for the type of end supports

= 1 for free end or hinged support

= 2.25 for fixed ends

= 1.60 for both ends pinned, guided and partially restrained

Torque in Nmm, Power; P in kW, speed; n in rpm

Eqn3.3 (a)/Pg.50

$$T = \frac{9.55 \times 10^6 (P)}{n}$$

Shaft design based on Rigidity

Shaft acts like a beam which deflects transversely and can deflect torsionally.

Torsional Rigidity

A shaft is said to be torsionally rigid if it doesn't twist too much under the action of external torque.

Examples: Cam shafts, spindles etc.

a. For solid shaft

Eqn3.2/Pg.50

$$\theta = \frac{TL}{GJ} \text{ rad} = \frac{584TL}{Gd^4} \text{ deg}$$

Where L = Length of the shaft, mm

T = Torque, Nmm

G = Modulus of Rigidity, N/mm²

J = Polar moment of Inertia, mm⁴

b. For Hollow shaft

$$\theta = \frac{584TL}{G(d_o^4 - d_i^4)} \text{ deg}$$

- The deflection should not exceed 1° in a length equal to twenty times the diameter of the shaft
- The permissible amount of twist for machine tool spindles, e.g. should not exceed 0.25° per meter length
- For line shafts 2.5° to 3° per meter length may be used as the limiting value

Lateral Rigidity

A shaft is said to be laterally rigid if it doesn't deflect too much under the action of external forces and bending moment.

It is important for maintain proper bearing clearances and for correct gear alignment. If the shaft is of uniform cross section, the deflection may be obtained from the deflection formulae as in solid mechanics. If it is of variable cross section the lateral deflection may be computed from either by graphical method is practical.

Double integration method to find out the deflection

$$\frac{d^2y}{d^2x} = \frac{M}{EI}$$

Evaluation of Transmission Forces

1. Forces due to belt drive

Let T_1 = Tension on tight side of the belt, N

T_2 = Tension on slack side of the belt, N

θ = Angle of contact and D = Diameter of the pulley, mm

Ratio of tensions $\frac{T_1}{T_2} = e^{\mu\theta}$ for flat belt Eqn14.3 (a)/Pg.290

$\frac{T_1}{T_2} = e^{\mu_1\theta}$ for V – belt Eqn14.12/Pg.294

Where μ = Coefficient of friction between the belt and the pulley

$\mu_1 = \frac{\mu}{\sin(\frac{\alpha}{2})}$ the apparent coefficient of friction, α = groove angle of pulley

- Effective or net pull, $F_t = (T_1 - T_2)$
- Torque $T = (T_1 - T_2)R$
- Total force or tension $= T_1 + T_2$

2. Forces due to Gear drive

- Tangential Force $F_t = \frac{2M_t}{d}$ Eqn14.8(a)/Pg.206
- Radial force $F_r = F_t \tan \alpha$ Eqn14.8(b)/Pg.206
 α = Pressure angle
- Pitch diameter $d = mz$, where m = module, z = number of teeth and M_t = transmitted torque, Nmm

Hollow Shaft

Compared with solid shaft, hollow shaft offers following advantages:

- (i) The stiffness of the hollow shaft is more than that of solid shaft with same weight.
- (ii) The strength of hollow shaft is more than that of solid shaft with same weight.
- (iii) The natural frequency of hollow shaft is higher than that of solid shaft with same weight.

Compared with solid shaft, hollow shaft has the following disadvantages:

(i) Hollow shaft is costlier than solid shaft.

(ii) The diameter of hollow shaft is more than that of solid shaft and requires more space

Hollow shafts are used to provide passage for coolants and control cables in case of deep hole drilling and bore well drilling. They are also used as propeller shafts in automobiles and for axles of railway wagons. Their applications include epicyclic gearboxes where one shaft rotates inside another. Hollow shafts are usually made by the extrusion process.

The design of hollow shaft consists of determining the correct inner and outer diameters from strength and rigidity considerations. Such shafts are subjected to axial tensile force, bending moment, torsional moment or combination of these loads.

Numericals

1. Compare the weight, strength and stiffness of a hollow shaft with that of a solid shaft having the same material and length.

2. A steel spindle transmits 4 kW at 800 rpm. The angular deflection should not exceed 0.25° per metre of the spindle. If the modulus of rigidity for the material of the spindle is 84 GPa, find the diameter of the spindle and the shear stress induced in the spindle.

Given: $P = 4 \text{ kW}$; $N = 800 \text{ rpm}$; $\theta = 0.25^\circ$; $L = 1 \text{ m} = 1000 \text{ mm}$; $G = 84 \text{ GPa} = 84 \times 10^9 \text{ N/m}^2 = 84 \times 10^3 \text{ N/mm}^2$

Torque in Nmm, Power; P in kW, speed; n in rpm

Eqn3.3 (a)/Pg.50

$$T = \frac{9.55 \times 10^6 (4)}{800} = \frac{9.55 \times 10^6 \times 4}{800} = 47750 \text{ Nmm}$$

For solid shaft

Eqn3.2/Pg.50

$$\theta = \frac{TL}{GJ} \text{ rad} = \frac{584TL}{Gd^4} \text{ deg}$$

$$0.25 = \frac{584 \times 47750 \times 1000}{84 \times 10^3 \times d^4}$$

Diameter = 33.9 mm

Nearest standard size = 36 mm, Table3.5 (a)/Pg.57

For solid shaft, shear stress ' τ '

$$\tau = \frac{16T}{\pi d^3} = \frac{16 \times 47750}{\pi \times 36^3} = 5.21 \frac{\text{N}}{\text{mm}^2}$$

3. Compare the shaft diameter on the basis of strength and torsional rigidity which is used to transmit 25 kW at 600 rpm. The permissible twist in the shaft is 0.5 deg in 2m length, $\tau_{\max} = 55 \text{ N/mm}^2$, $G = 80 \text{ GPa}$

Given: Power = 25 kW; speed = 600 rpm, Permissible twist = 0.5 deg; Length = 2m; $\tau_{\max} = 55 \text{ N/mm}^2$; $G = 80 \text{ GPa} = 80 \times 10^3 \text{ N/mm}^2$

Design based on strength

Torque in Nmm, Power; P in kW, speed; n in rpm

Eqn3.3 (a)/Pg.50

$$T = \frac{9.55 \times 10^6 (P)}{n} = \frac{9.55 \times 10^6 \times 25}{600} = 397917 \text{ Nmm}$$

For solid shaft, shear stress ' τ '

$$\tau = \frac{16T}{\pi d^3}$$

$$d = \sqrt[3]{\frac{16T}{\pi \tau_{\max}}} = \sqrt[3]{\frac{16 \times 397917}{\pi \times 55}} = 33.27 \text{ mm} \sim 34 \text{ mm}$$

Design based on torsional rigidity

For solid shaft

Eqn3.2/Pg.50

$$\theta = \frac{TL}{GJ} \text{ rad} = \frac{584TL}{Gd^4} \text{ deg}$$

$$0.5 = \frac{584 \times 397917 \times 2000}{80 \times 10^3 \times d^4}$$

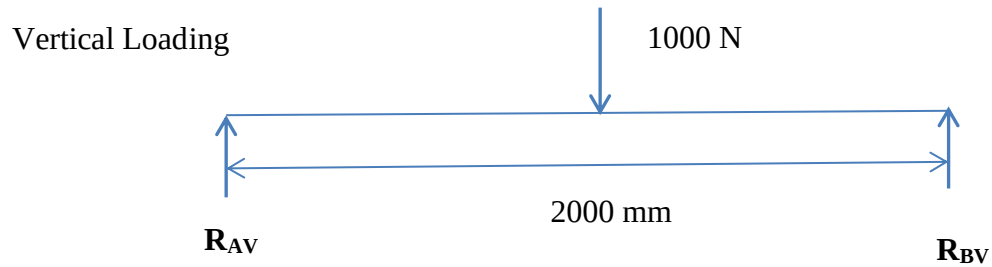
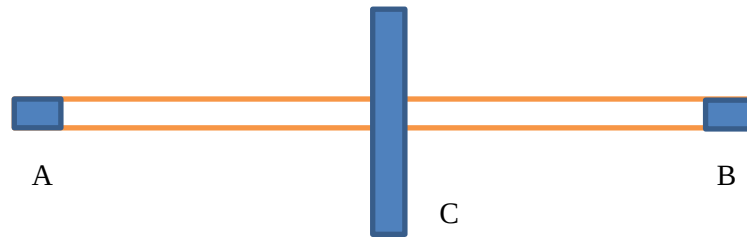
$$d = 58.38 \text{ mm} \sim 60 \text{ mm}$$

Compare to the two values, the shaft diameter = 60 mm

4. A section of commercial shafting 2m long between bearings carries a 1000 N pulley at its midpoint. The pulley is keyed to the shaft and receives 30 kW at 150 rpm which is transmitted to a flexible coupling just outside the right bearing. The belt drive is horizontal and the sum of the belt tensions is 8000 N. Assume $C_m = C_t = 1.5$ Calculate the shaft size.

Given: Length of the shaft = 2m = 2000 mm, Weight of the pulley = 1000N (Vertical), Power P = 30 kW, Speed n = 150 rpm, Sum of belt tensions = 8000 N (Horizontal), $C_m = C_t = 1.5$.
Torque in Nmm, Power; P in kW, speed; n in rpm Eqn3.3 (a)/Pg.50

$$T = \frac{9.55 \times 10^6 (P)}{n} = \frac{9.55 \times 10^6 \times 30}{150} = 1.91 \times 10^6 \text{ Nmm}$$



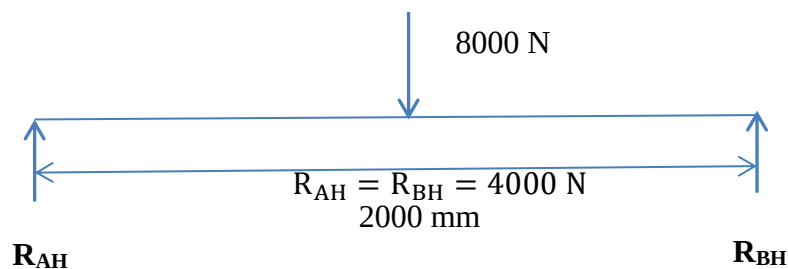
$$R_{AV} = R_{BV} = 500 \text{ N}$$

- Bending Moments

$$M_{AV} = M_{BV} = 0$$

$$M_{CV} = R_{AV} \times 1000 = 500 \times 1000 = 500 \times 10^3 \text{ Nmm}$$

Horizontal Loading



- Bending Moments

$$M_{AH} = M_{BH} = 0$$

$$M_{CH} = R_{AH} \times 1000 = 4000 \times 1000 = 4000 \times 10^3 \text{ Nmm}$$

- Resultant Bending Moments

Resultant bending moment at A and B = 0

Resultant bending moment at C

$$M_C = \sqrt{(M_{CV})^2 + (M_{CH})^2} = \sqrt{(500 \times 10^3)^2 + (4000 \times 10^3)^2} = 4031.13 \times 10^3 \text{ Nmm}$$

➤ Maximum shear stress theory

Eqn3.6 (b)/Pg.51

$$d_o = \left[\frac{16}{\pi \tau_{\max}} \left(\sqrt{(C_m M)^2 + (C_t T)^2} \right) \times \left(\frac{1}{1 - K^4} \right) \right]^{\frac{1}{3}}$$

For Solid Shaft, $K = 0$, Commercial steel shafting with allowance for keyways

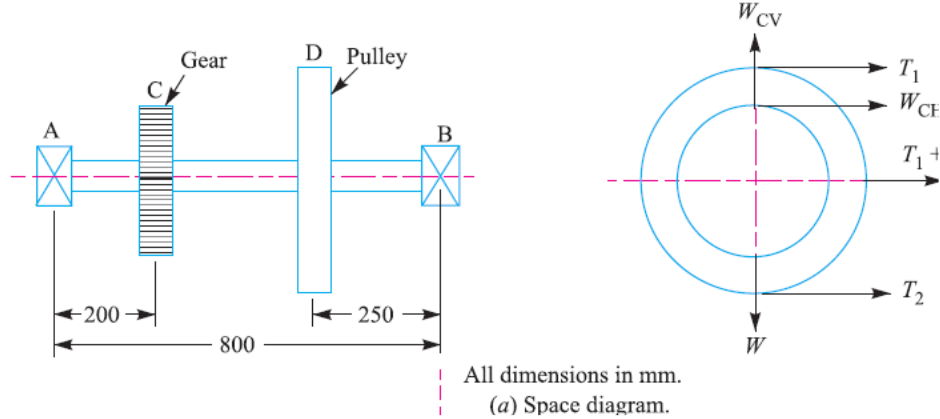
$\tau_{\max} = 41 \text{ N/mm}^2$ Table3.5(b)/Pg.57

, $C_m = C_t = 1.5$

$$\begin{aligned} d_o = d &= \left[\frac{16}{\pi \tau_{\max}} \left(\sqrt{(C_m M)^2 + (C_t T)^2} \right) \right]^{\frac{1}{3}} \\ &= \left[\frac{16}{\pi \times 41} \left(\sqrt{(1.5 \times 4031.13 \times 10^3)^2 + (1.5 \times 1.91 \times 10^6)^2} \right) \right]^{\frac{1}{3}} \end{aligned}$$

Diameter of the shaft = 94.02 mm, Nearest standard size = 100 mm, Table3.5 (a)/Pg.57

5. A shaft is supported on bearings A and B, 800 mm between centres. A 20° straight tooth spur gear having 600 mm pitch diameter, is located 200 mm to the right of the left hand bearing A, and a 700 mm diameter pulley is mounted 250 mm towards the left of bearing B. The gear is driven by a pinion with a downward tangential force while the pulley drives a horizontal belt having 180° angle of wrap. The pulley also serves as a flywheel and weighs 2000 N. The maximum belt tension is 3000 N and the tension ratio is 3: 1. Determine the maximum bending moment and the necessary shaft diameter if the allowable shear stress of the material is 40 MPa.



Given:

$AB = 800 \text{ mm}$; $\alpha = 20^\circ$; $D_C = 600 \text{ mm}$; $AC = 200 \text{ mm}$; $D_D = 700 \text{ mm}$; $DB = 250 \text{ mm}$; $\theta = 180^\circ$;
 $W = 2000 \text{ N}$; $T_1 = 3000 \text{ N}$; $T_1/T_2 = 3$; $\tau = 40 \text{ MPa} = 40 \text{ N/mm}^2$

Power and speed is not given

- Pulley and Belt at D
- Torque $T = (T_1 - T_2)R_D = T_1 \left(1 - \frac{T_2}{T_1}\right) R_D = 3000 \left(1 - \frac{1}{3}\right) \frac{700}{2} = 700 \times 10^3 \text{ Nmm}$
- Belt tensions
 $T_1 = 3000 \text{ N}$; $T_1/T_2 = 3$, $T_2 = 1000 \text{ N}$
 Total Tensions $T_1 + T_2 = 3000 + 1000 = 4000 \text{ N}$ (Horizontal)

Weight of the Pulley = $W_p = 2000 \text{ N}$ (Vertical)

- Gear at C

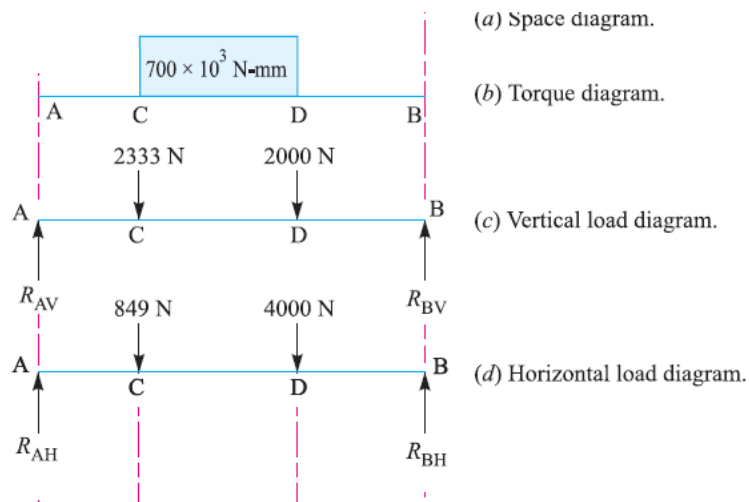
Assuming the torque at C = torque at D = $700 \times 10^3 \text{ Nmm}$

Tangential force acting in the gear

$$F_{tC} = \frac{2M_t}{D_C} = \frac{2 \times 700 \times 10^3}{600} = 2333 \text{ N (Vertical)}$$

Radial Force

$$F_{rC} = F_{tC} \tan \alpha = 2333 \times \tan 20 = 849 \text{ N (Horizontal)}$$



Vertical Loading

$$R_{AV} + R_{BV} = 2333 + 2000 = 4333 \text{ N}$$

Taking moment at B

$$R_{AV} \times 800 = 2333 \times 600 + 2000 \times 250$$

$$R_{AV} = 2375 \text{ N}$$

$$R_{BV} = 4333 - 2375 = 1958 \text{ N}$$

- Bending Moments

$$M_{AV} = M_{BV} = 0$$

$$M_{CV} = R_{AV} \times 200 = 2375 \times 200 = 475000 \text{ Nmm}$$

$$M_{DV} = R_{BV} \times 250 = 1958 \times 250 = 489500 \text{ Nmm}$$

Horizontal Loading

$$R_{AH} + R_{BH} = 849 + 4000 = 4849 \text{ N}$$

Taking moment at B

$$R_{AH} \times 800 = 849 \times 600 + 4000 \times 250$$

$$R_{AH} = 1887 \text{ N}$$

$$R_{BH} = 4849 - 1887 = 2962 \text{ N}$$

- Bending Moments

$$M_{AH} = M_{BH} = 0$$

$$M_{CH} = R_{AH} \times 200 = 1887 \times 200 = 377400 \text{ Nmm}$$

$$M_{DH} = R_{BH} \times 250 = 2962 \times 250 = 740500 \text{ Nmm}$$

- Resultant Bending Moments

Resultant bending moment at A and B = 0

Resultant bending moment at C

$$M_C = \sqrt{(M_{CV})^2 + (M_{CH})^2} = \sqrt{(475000)^2 + (377400)^2} = 606676 \text{ Nmm}$$

Resultant bending moment at D

$$M_D = \sqrt{(M_{DV})^2 + (M_{DH})^2} = \sqrt{(489500)^2 + (740500)^2} = 887666 \text{ Nmm}$$

Maximum Bending Moment is at D = 887666 Nmm

Maximum shear stress theory

Eqn3.6 (b)/Pg.51

$$d_o = \left[\frac{16}{\pi \tau_{\max}} \left(\sqrt{(C_m M)^2 + (C_t T)^2} \right) \times \left(\frac{1}{1 - K^4} \right) \right]^{\frac{1}{3}}$$

For Solid Shaft, K = 0

$$d_o = \left[\frac{16}{\pi \tau_{\max}} \left(\sqrt{(C_m M)^2 + (C_t T)^2} \right) \right]^{\frac{1}{3}}$$

Assuming steady or gradually applied load for rotating shaft, shock and endurance factors for bending = $C_m = 1.5$ and for torsion $C_t = 1$ (Table 3.1/Pg.56)

$$d_o = d = \left[\frac{16}{\pi \times 40} \left(\sqrt{(1.5 \times 887666)^2 + (1 \times 700 \times 10^3)^2} \right) \right]^{\frac{1}{3}} = 57.6 \text{ mm}$$

Standard size (Table 3.5(a)/Pg.57)

Diameter of the shaft = 63 mm

6. A machine shaft turning at 600 rpm is supported on bearings 750 mm apart. 15 kW is supplied through a 450 mm pulley located at 250 mm to the right of the right bearing. The power is

transmitted from the shaft through a 200 mm gear located 250 mm to the right of the left bearing. The belt drive is at an angle of 60° above horizontal. The pulley weighs 800 N to provide some flywheel effect. The ratio of belt tensions is 3:1. The gear has 20° tooth form and mates with another gear located directly above the shaft. If the shaft material selected to have an ultimate tensile strength of 500 MPa and yield strength of 310 MPa, determine the required diameter of the shaft.

Given: Length of the shaft = 750+250 = 1000 mm, Diameter of the pulley = 450 mm, Weight of the pulley = 800N (Vertical), Power P = 15 kW, Speed n = 600 rpm, Pitch circle diameter of gear = 200 mm, $T_1/T_2 = 3$ Sum of belt tensions = T_1+T_2 N (60° above horizontal), $C_m = 1.5$, $C_t = 1.0$, $\sigma_u = 500$ MPa, $\tau = 310$ MPa

Torque in Nmm, Power; P in kW, speed; n in rpm

Eqn3.3 (a)/Pg.50

$$T = \frac{9.55 \times 10^6 (P)}{n} = \frac{9.55 \times 10^6 \times 15}{600} = 238.75 \times 10^3 \text{ Nmm}$$

Pulley and Belt

- Torque $T = 238.75 \times 10^3 = (T_1 - T_2)R_p = T_1 \left(1 - \frac{T_2}{T_1}\right) \frac{450}{2} = T_1 \left(1 - \frac{1}{3}\right) 225$

- Belt tensions

$$T_1 = 1591.7 \text{ N}; T_1/T_2 = 3, T_2 = 530.6 \text{ N}$$

$$\text{Total Tensions } T_1 + T_2 = 1591.7 + 530.6 = 2122.3 \text{ N (60° above Horizontal)}$$

$$\text{Horizontal component of belt tensions} = 2122.3 \cos 60 = 1061.15 \text{ N}$$

$$\text{Vertical component of belt tensions} = 2122.3 \sin 60 = 1838 \text{ N (Upwards)}$$

$$\text{Weight of the Pulley} = W_p = 800 \text{ N (Vertically downwards)}$$

$$\text{Net upward Force} = 1838 - 800 = 1038 \text{ N}$$

Gear

Assuming the torque in the shaft = torque in gear = 238.75×10^3 Nmm

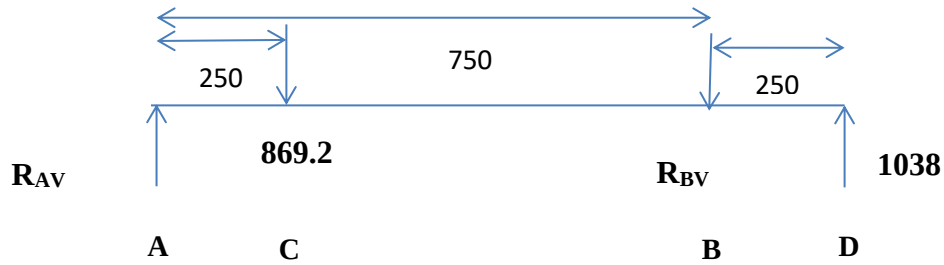
Tangential force acting in the gear

$$F_t = \frac{2M_t}{200} = \frac{2 \times 238.75 \times 10^3}{200} = 2388 \text{ N (Horizontal)}$$

Radial Force

$$F_r = F_t \tan \alpha = 2388 \times \tan 20 = 869.2 \text{ N (Vertical)}$$

Vertical Loading



Taking moments at the left end

$$R_{BV} \times 750 = 1038 \times 1000 - 869.2 \times 250$$

$$R_{BV} = 1094.3 \text{ N}$$

$$R_{AV} = 1094.3 + 869.2 - 1038 = 925.5 \text{ N}$$

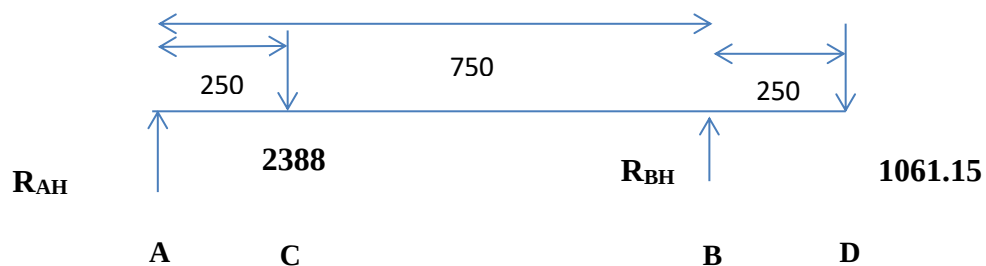
- Bending Moments

$$M_{AV} = M_{DV} = 0$$

$$M_{CV} = R_{AV} \times 250 = 925.5 \times 250 = 231375 \text{ Nmm}$$

$$M_{BV} = 1038 \times 250 = 259500 \text{ Nmm}$$

Horizontal Loading



Taking moments at the left end

$$R_{BH} \times 750 = 1061.15 \times 1000 + 2388 \times 250$$

$$R_{BH} = 2211 \text{ N}$$

$$R_{AH} = 2388 + 1061.15 - 2211 = 1238.15 \text{ N}$$

- Bending Moments

$$M_{AH} = M_{DH} = 0$$

$$M_{CH} = R_{AH} \times 250 = 1238.15 \times 250 = 309538 \text{ Nmm}$$

$$M_{BH} = 1061.15 \times 250 = 265288 \text{ Nmm}$$

- Resultant Bending Moments

Resultant bending moment at A and D = 0

Resultant bending moment at C

$$M_C = \sqrt{(M_{CV})^2 + (M_{CH})^2} = \sqrt{(231375)^2 + (309538)^2} = 386456 \text{ Nmm}$$

Resultant bending moment at B

$$M_B = \sqrt{(M_{BV})^2 + (M_{BH})^2} = \sqrt{(259500)^2 + (265288)^2} = 371104 \text{ Nmm}$$

Maximum Bending Moment is at C = 386456 Nmm

Maximum permissible working stresses for shafts

Table 3.5 (a)/Pg.57

$$\tau_{\max} = 0.3\sigma_y = 0.3 \times 310 = 93 \text{ MPa}$$

$$\tau_{\max} = 0.18\sigma_u = 0.18 \times 500 = 90 \text{ MPa}$$

Since keyways are present, the values are to be reduced by 25 %

$$\text{Allowable shear stress} = 0.75 \times 90 = 67.5 \text{ MPa} = 67.5 \text{ N/mm}^2$$

Maximum shear stress theory

Eqn3.6 (b)/Pg.51

$$d_o = \left[\frac{16}{\pi \tau_{\max}} \left(\sqrt{(C_m M)^2 + (C_t T)^2} \right) \times \left(\frac{1}{1 - K^4} \right) \right]^{\frac{1}{3}}$$

For Solid Shaft, K = 0

$C_m = 1.5$ and $C_t = 1$

$$d_o = d = \left[\frac{16}{\pi \tau_{\max}} \left(\sqrt{(C_m M)^2 + (C_t T)^2} \right) \right]^{\frac{1}{3}}$$

$$d = \left[\frac{16}{\pi \times 67.5} \left(\sqrt{(1.5 \times 386456)^2 + (1 \times 238.75 \times 10^3)^2} \right) \right]^{\frac{1}{3}} = 36.2 \text{ mm}$$

Standard size (Table 3.5(a)/Pg.57)

Diameter of the shaft = 40 mm

7. A solid steel shaft is supported on two bearings 1.8 m apart and rotates at 250 r.p.m. A 20° involute gear D, 300 mm diameter is keyed to the shaft at a distance of 150 mm to the left on the right hand bearing. Two pulleys B and C are located on the shaft at distances of 600 mm and 1350 mm respectively to the right of the left hand bearing. The diameters of the pulleys B and C are 750 mm and 600 mm respectively. 30 kW is supplied to the gear, out of which 18.75 kW is taken off at the pulley C and 11.25 kW from pulley B. The drive from B is vertically downward while from C the drive is downward at an angle of 60° to the horizontal. In both cases the belt tension ratio is 2 and the angle of lap is 180°. The combined fatigue and shock factors for torsion and bending may be taken as 1.5 and 2 respectively. Design a suitable shaft taking working stress to be 42 MPa in shear and 84 MPa in tension.

8. A hollow shaft 500 mm outside diameter and 300 mm inside diameter is supported by two bearings 6 m apart. The shaft is driven by a flexible coupling at one end and drives a ship's propeller at 100 rpm. The maximum thrust on the propeller is 500 kN when the shaft is transmitting 6000 kW. The shaft weighs 60 kN. Determine the maximum shear stress induced in the shaft. The combined fatigue and shock factors for bending and torsion may be taken as 1.5 and 1 respectively.

Given: $d_o = 500$ mm; $d_i = 300$ mm, $L = 6$ m = 6000 mm; rpm = 100 rpm; $F = 500$ kN = 500×10^3 N; Power = 6000 kW; $W = 60$ kN = 60×10^3 N; $C_m = 1.5$; $C_t = 1$

Torque in Nmm, Power; P in kW, speed; n in rpm

Eqn3.3 (a)/Pg.50

$$T = \frac{9.55 \times 10^6 (P)}{n} = \frac{9.55 \times 10^6 \times 6000}{100} = 573 \times 10^6 \text{ Nmm}$$

Consider the shaft as simply supported beam with uniform load, maximum bending Moment
Table 1.4/Pg.16, 3rd column, 6th row

$$M_{\max} = \frac{WL}{8} = \frac{60000 \times 6000}{8} = 45 \times 10^6 \text{ Nmm}$$

Calculation of L/k where k = radius of gyration. For hollow shaft, as per the table 1.3 (a)/Pg.13 6th column, row 'h'

$$k = \frac{\sqrt{D_1^2 + D_2^2}}{4} = \frac{\sqrt{500^2 + 300^2}}{4} = 146 \text{ mm}$$

$$\frac{L}{k} = \frac{6000}{146} = 41.1 \text{ which is } < 115$$

α = ratio of maximum intensity of stress resulting from the axial load to the average axial stress

$$\alpha = \frac{1}{1 - 0.0044 \left(\frac{L}{k} \right)} \quad \text{When } L/k < 115$$

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$$\alpha = \frac{1}{1-0.0044(41.1)} = 1.22$$

Ratio of diameters

$$K = \frac{d_i}{d_o} = \frac{300}{500} = 0.6$$

The outside diameter of hollow shaft subjected to an axial load F in addition to the bending and torsional loads. Eqn3.8 (b)/Pg.51

$$C_m = 1.5, C_t = 1, F = F = 500 \text{ kN} = 500 \times 10^3 \text{ N}$$

$$d_o = \left[\frac{16}{\pi \tau_{\max}} \left(\sqrt{\left(C_m M + \frac{\alpha F d_o (1 + K^2)}{8} \right)^2 + (C_t T)^2} \right) \times \left(\frac{1}{1 - K^4} \right) \right]^{\frac{1}{3}}$$

$$d_o^3 = \frac{16}{\pi \tau_{\max}} \left(\sqrt{\left(C_m M + \frac{\alpha F d_o (1 + K^2)}{8} \right)^2 + (C_t T)^2} \right) \times \left(\frac{1}{1 - K^4} \right)$$

$$500^3 = \frac{16}{\pi \times \tau_{\max}} \left(\sqrt{\left(1.5 \times 45 \times 10^6 + \frac{1.22 \times 500 \times 10^3 \times 500 (1 + 0.6^2)}{8} \right)^2 + (1 \times 573 \times 10^6)^2} \right) \times \left(\frac{1}{1 - 0.6^4} \right)$$

Maximum shear stress

$$\tau_{\max} = 27.4 \frac{\text{N}}{\text{mm}^2}$$

9. Prove that the a hollow shaft is always stronger than a solid shaft made of same material, length and weight when subjected to simple torsion

Given: Material same – Specific weight ‘w’, Allowable shear stress and Modulus of rigidity are same both for solid and hollow shafts.

Length is same ‘l’, Weight is same

$$W_s = W_H$$

$$\left(\frac{\pi}{4} d^2 \right) l w = \frac{\pi}{4} (d_o^2 - d_i^2) l w$$

$$d^2 = d_o^2 - d_i^2 = d_o^2 \left(1 - \frac{d_i^2}{d_o^2} \right) = d_o^2 (1 - K^2)$$

$$d = d_o \sqrt{1 - K^2}$$

Shear stress

$$\frac{T}{J} = \frac{\tau}{r}$$

$$\tau = \frac{T \times r}{J}$$

For solid shaft

$$\tau_s = \frac{T_s \times r_s}{\frac{\pi}{32} d^4} = \frac{T_s \times (d/2)}{\frac{\pi}{32} d^4} = \frac{16T_s}{\pi d^3}$$

For hollow shaft

$$\tau_H = \frac{T_H \times r_H}{\frac{\pi}{32} (d_o^4 - d_i^4)} = \frac{T_s \times (d_o/2)}{\frac{\pi}{32} d_o^4 (1 - K^4)} = \frac{16T_H}{\pi d_o^3 (1 - K^4)}$$

$$\tau_H = \tau_s$$

$$\frac{16T_H}{\pi d_o^3 (1 - K^4)} = \frac{16T_s}{\pi d^3}$$

$$\frac{T_H}{T_s} = \frac{d_o^3 (1 - K^4)}{d^3}$$

$$d = d_o \sqrt{1 - K^2}$$

$$\frac{T_H}{T_s} = \frac{d_o^3 (1 - K^4)}{(d_o \sqrt{1 - K^2})^3} = \frac{d_o^3 (1 + K^2)(1 - K^2)}{d_o^3 (1 - K^2) \sqrt{1 - K^2}} = \frac{(1 + K^2)}{\sqrt{1 - K^2}}$$

This is always >1 as $K < 1$ and $1+K > 1 - K$.

Hence hollow shaft is always stronger than a solid shaft made of same material, length and weight when subjected to simple torsion